

ORIGINAL REPORT OPEN ACCESS

Novel Injectable Supports Long-Term Recurrent Laryngeal Nerve and Partial Peripheral Nerve Repair

Aditya Bhatt¹  | William Boswell¹ | Elena Kennedy^{1,2} | Sunjay Anekal¹  | Emma Mazurek^{1,2} | Ananya Tadikonda^{1,2}  | Christopher Rafalko¹ | Jasmine Strelow³ | Isaac Shipchandler⁴ | Patrick R. Finnegan^{1,2} | Stacey Halum^{1,2}

¹Indiana University School of Medicine (IUSM), Indianapolis, Indiana, USA | ²Indiana University School of Medicine Department of Otolaryngology-Head and Neck Surgery, Indianapolis, Indiana, USA | ³University of Wisconsin-Madison, Madison, Wisconsin, USA | ⁴Indiana University, Bloomington, Indiana, USA

Correspondence: Stacey Halum (shalum@iu.edu)

Received: 30 March 2026 | **Revised:** 21 July 2026 | **Accepted:** 4 August 2026

Keywords: nerve regeneration | recurrent laryngeal nerve | vocal fold paralysis

ABSTRACT

Background: Unilateral vocal fold paralysis (VFP) from recurrent laryngeal nerve (RLN) injury currently has no injectable treatment options that promote targeted reinnervation. In a prior pilot study, Agrin, neuregulin-1 (NRG1), and acetylcholine (ACh) enhanced early (1-month) reinnervation after RLN injury. The aims of the current study were to (1) assess if these effects persist in a 3-month RLN injury model and (2) test whether Agrin/ACh/NRG1 promotes regeneration after sciatic nerve (SN) transection, with relevance to nerve repair beyond the larynx.

Methods: Adult C57BL/6 mice underwent unilateral RLN or SN transection with injections into the thyroarytenoid or gastrocnemius, respectively. Separate RLN and SN cohorts included saline ($N=5$), biomolecule (Agrin, NRG1, ACh) ($N=5$), and uninjured controls ($N=5$). Functional recovery was evaluated using videolaryngoscopy and electromyography (EMG) for RLN injury, and gait analysis with EMG for the SN model. Histologic outcomes included immunohistochemistry and myofiber diameter of the thyroarytenoid or gastrocnemius muscles.

Results: Biomolecule treatment improved vocal fold movement and restored EMG symmetry in the RLN model, consistent with functional reinnervation. Histology demonstrated increased NMJ formation and preserved myofiber integrity. In the SN model, treatment did not produce meaningful functional recovery based on EMG and gait deficits. However, gross nerve continuity, increased neurofilament staining, and preserved myofiber architecture indicated partial structural regeneration.

Conclusions: Biomolecule therapy supports functional and structural regeneration following RLN injury and demonstrates structural, but not sustained functional, recovery in the SN model. Further investigation into sustained delivery or repeat dosing may be necessary to support regeneration in larger peripheral nerves.

Level of Evidence: N/A.

1 | Introduction

Recurrent laryngeal nerve (RLN) injury is a major cause of unilateral vocal fold paralysis (VFP), and no minimally invasive

treatment reliably restores adductor motion [1–7]. Current surgical interventions primarily improve glottic closure rather than re-establishing physiologic neuromuscular junction (NMJ) architecture [8–10]. Reinnervation techniques aim to restore

This study was presented at the American Broncho-Esophagological Association (ABEA) Annual Meeting; April 22–26, 2026; Phoenix, AZ.

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *The Laryngoscope* published by Wiley Periodicals LLC on behalf of The American Laryngological, Rhinological and Otolological Society, Inc.

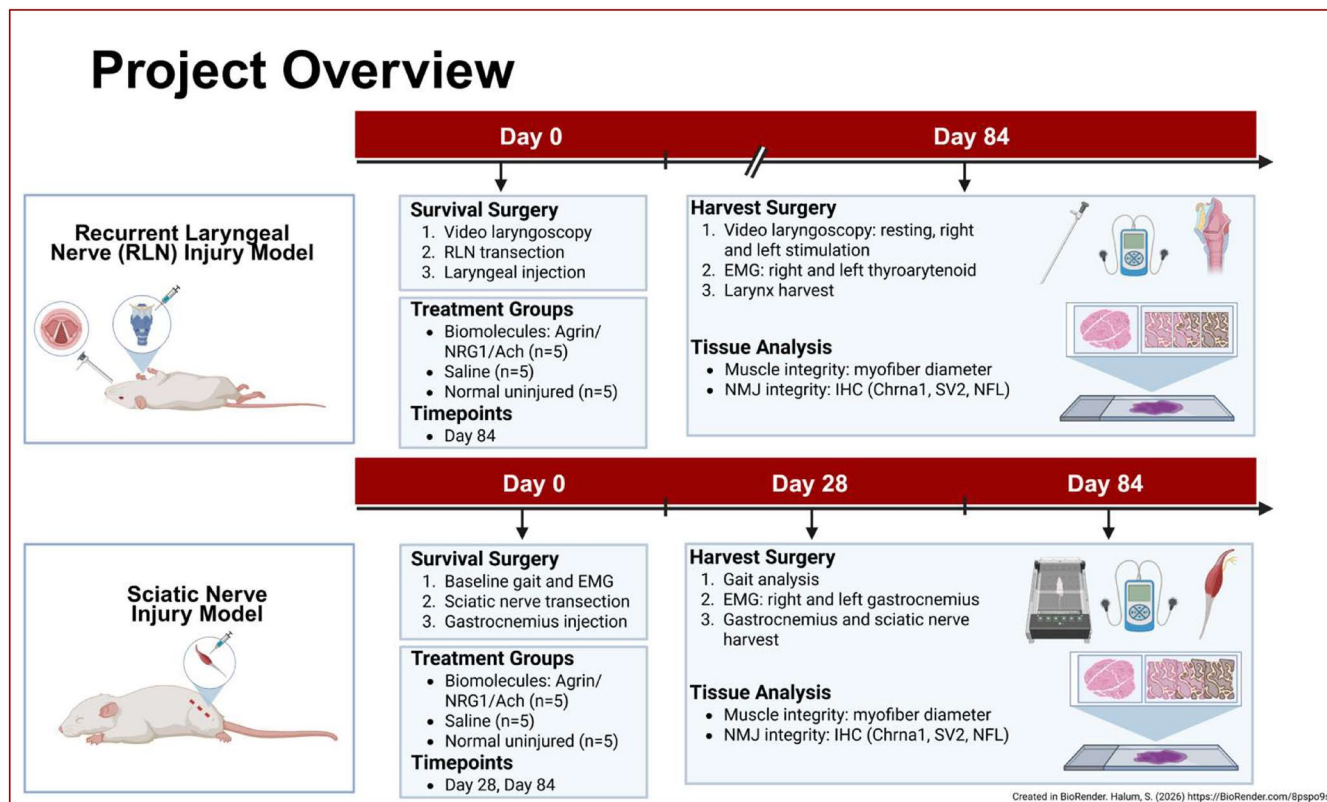


FIGURE 1 | Project overview demonstrating experimental timelines for RLN and SN injury models. Animals underwent nerve transection and treatment followed by longitudinal functional and histologic assessment to evaluate neuromuscular regeneration and reinnervation.

adduction but are constrained by synkinesis and incomplete recovery [7, 11–12]. These limitations highlight the need for biologic strategies that promote neuromuscular regeneration rather than static medialization.

Laryngeal denervation triggers axonal and synaptic remodeling: distal axons undergo Wallerian degeneration and Schwann cells guide regrowth toward target fibers [13–18]. Loss of neural input fragments the postsynaptic apparatus and disperses acetylcholine (ACh) receptors [19, 20]. Regenerating axons lack original topographic cues, so reinnervation is often incomplete or misdirected, producing synkinesis and chronic impairment [21–23].

Agrin–MuSK signaling clusters ACh receptors at the postsynaptic membrane, whereas neuregulin-1 (NRG1)–ErbB3/4 signaling regulates receptor transcription and Schwann cell-mediated axonal repair and guidance [24–28]. Activity-dependent ACh signaling stabilizes synapses [19, 20]. After nerve injury, these pathways show transient activation that is not sustained, making them attractive targets for enhancing NMJ formation and muscle reinnervation by maintaining receptive motor endplates and reparative Schwann cells [29–33].

Prior work showed that co-injection of Agrin, NRG1, and ACh into denervated thyroarytenoid (TA) muscle enhances early (1-month) structural and electrophysiologic reinnervation after RLN transection [34]. Because synaptic maturation and synkinetic pathways evolve over months [21, 35], this study extended follow-up to 3 months to assess RLN outcomes in a murine model. We also asked whether this biomolecular strategy

generalizes beyond the larynx using a sciatic nerve (SN) transection model, a well-characterized system for testing peripheral nerve regeneration with growth factors, Schwann cell scaffolds, and other biologics [36–40].

This study aimed to determine whether localized Agrin/ NRG1/ACh delivery produces sustained structural and functional recovery of VF adduction in a three-month RLN injury model and whether this biomolecular strategy promotes peripheral nerve regeneration. We hypothesized that targeted modulation of NMJ-associated signaling would stabilize NMJs and enhance axonal regeneration after RLN and SN injury in mice. Figure 1 summarizes the injury models, timeline, and treatment groups.

2 | Materials and Methods

2.1 | Injection Preparation

Biomolecule treatment consisted of 500 nM agrin (Catalog# 6624-AG, R&D Systems, Minneapolis, Minnesota), 1 nM NRG1 (Catalog# 5898-NR, R&D Systems, Minneapolis, Minnesota), and 100 μ M ACh chloride (Catalog# 159170050, Thermo Scientific, Fair Lawn, New Jersey) in sterile 0.9% saline, selected based on established NMJ roles and prior work showing this combination promotes motor endplate formation and reinnervation after RLN injury [14–18, 35]. Concentrations exceeded in vitro effective ranges to account for diffusion and clearance after intramuscular delivery.

Solutions were prepared 1 week before surgery, aliquoted, stored at -20°C , and used with identical formulations in RLN and sciatic experiments. Detailed preparation protocols and reagent information are provided in Data S1.

2.2 | Survival Surgery

All procedures were performed in accordance with an IACUC-approved protocol (23150) and followed ARRIVE guidelines for animal research. Male C57BL/6 mice were selected due to their wide use in RLN injury and peripheral nerve regeneration models [35–41]. Separate cohorts of animals were used for RLN and SN experiments. Within each injury model, mice were allocated to one of three groups: (1) saline control, (2) biomolecule treatment, and (3) uninjured control. **RLN animals were harvested at 84 days**, while SN animals were harvested at 28 or 84 days following injury (see Table S1).

For the RLN model, mice in the saline ($n=5$) and biomolecule ($n=5$) groups underwent survival surgery in which the right RLN was exposed through a ventral cervical approach and transected at the level of the fourth to sixth tracheal ring. Vocal fold (VF) immobility was confirmed via video laryngoscopy. **A single $10\mu\text{L}$ injection was delivered transcartilaginously into the right TA muscle** through the center of the right thyroid cartilage by the same investigator using a standardized approach. This volume was selected based on prior work from our group demonstrating reliable intramuscular distribution within the murine TA without paralaryngeal extravasation [35], and confirmed with pilot dye-injection testing. Single, immediate treatment administration was chosen to target the early regenerative window following injury during which NMJ-associated signaling pathways are most receptive to augmentation [19–21]. Lastly, the proximal and distal nerve stumps were reapproximated without direct repair.

In the SN model, the left SN was exposed through a posterior thigh incision and transected proximal to the trifurcation. Biomolecule treatment was delivered to the injured limb via intramuscular gastrocnemius injection and topical application to the transected nerve; the contralateral gastrocnemius served as within-animal control. Injection volumes were scaled to muscle size and refined by pilot testing to maximize intramuscular distribution while minimizing tissue distension (see Data S1).

2.3 | Laryngeal Testing

Transoral videolaryngoscopy (Ambu aScope 5 Broncho 2.7/1.2 mm, Catalog# 624001000US, Ambu, Columbia, Maryland) was performed to assess VF motion and positioning before and after RLN transection. At harvest, baseline spontaneous VF movement was assessed, followed by independent stimulation of the right and left RLNs to evaluate stimulation-induced VF twitching as a measure of functional reinnervation. All videolaryngoscopy recordings were de-identified by an independent investigator and evaluated in a blinded fashion without knowledge of treatment allocation.

2.4 | Electromyography

EMG was performed at harvest in both models using the UltraPro S100 EMG system (Catalog# 828-060800, Natus, Middleton, Wisconsin). For the RLN model, the right and left RLNs were stimulated independently at 5.10 mA and 1 Hz, with recordings obtained from the ipsilateral TA muscle, the primary VF adductor and site of biomolecule delivery. Area under the curve (AUC) values were calculated for quantitative comparison between treatment groups. For the SN model, baseline EMG was recorded prior to transection and repeated at harvest for the injured left and contralateral right hindlimb. The SN was stimulated at 1.18 mA and 1 Hz, with evoked responses recorded from the gastrocnemius muscle and AUC values calculated for each recording.

2.5 | Sciatic Functional Assessment

Gait was assessed using the Kinetic Weight Bearing system (BIO-KWB-DUAL, Bioseb, Vitrolles, France). Limb-specific variables including peak force, force curve area (FCA), stance duration, propulsion duration, and stride length were collected for each hindlimb. Injured left limb values were compared to contralateral right limb values to quantify interlimb asymmetry (for additional details, see Data S1).

2.6 | Histology and Immunohistochemistry

For RLN analysis, larynges were fixed, embedded, and sectioned coronally for TA muscle assessment. Because the TA runs anteroposteriorly and slightly laterally, coronal sections of the trachea contain a mix of transverse, longitudinal, and oblique fibers; myofiber size was therefore quantified using the minimum Feret's diameter (minFeret), which is minimally sensitive to sectioning angle and fiber orientation. Immunohistochemical staining was performed to evaluate NMJ and nerve regeneration using antibodies against ACh receptors (CHRNA1; Catalog# NBP2-81070R, Novus Biologicals, Centennial, Colorado), synaptic vesicle protein 2 (SV2; Catalog# ABIN916918, Antibodies Online, Aachen, Germany), and neurofilament light chain (NFL; Catalog# ab307394, Abcam, Waltham, Massachusetts).

For SN analysis, gastrocnemius muscles were fixed, embedded, and sectioned in cross-sectional and longitudinal planes, and axonal regeneration assessed by NFL staining of nerve fibers. Myofiber diameter in both models was quantified from digitized H&E cross-sections in FIJI using watershed-based segmentation, calculating minimum Feret's diameter per fiber and averaging per specimen (see Figure S1 and Data S1 for staining protocols and myofiber-analysis parameters).

2.7 | Statistical Analysis

All analyses were performed in R (R Foundation for Statistical Computing, Vienna, Austria), with significance defined as $\alpha=0.05$. Test selection was guided by data structure: paired

versus independent samples, repeated measures, ordinal versus continuous outcomes, and variance homogeneity.

2.7.1 | Electromyography

Within each treatment group, injured-side AUC was compared to the contralateral uninjured side using a paired *t*-test, reflecting the paired within-animal design. Between-group recovery was compared using independent *t*-tests across separate treatment cohorts. The same framework was applied to both RLN and SN EMG data.

2.7.2 | Gait and Weight-Bearing

Longitudinal gait parameters were analyzed with linear mixed-effects models (treatment, time, and treatment $\times$ time as fixed effects; random intercept per animal) to account for repeated measurements within animals, followed by Tukey-adjusted pairwise post hoc comparisons. Limb symmetry was assessed by one-sample *t*-test of the injured-to-uninjured force-curve area ratio against the reference value of 1.0 (perfect symmetry).

2.7.3 | Video Laryngoscopy

VF motion ordinal positioning scores were analyzed using the Kruskal-Wallis test with Dunn post hoc comparisons and Benjamini-Hochberg correction.

2.7.4 | Histologic Quantification

Myofiber minFeret diameters were continuous. Normality was confirmed by the Shapiro-Wilk test, and groups were compared using Welch's ANOVA with Games-Howell post hoc analysis, chosen for robustness to unequal variances across groups.

3 | Results

3.1 | Functional Outcomes

3.1.1 | Video Laryngoscopy

Right RLN stimulation resulted in a right VF response in 4/5 biomolecule mice compared to 0/5 saline mice ($p=0.019$), while all 5/5 uninjured mice responded. Left RLN stimulation responses were intact across all groups (5/5), confirming injury specificity to the right RLN. At 84 days, 3/5 biomolecule mice demonstrated midline VF positioning compared to 0/5 saline mice, which showed predominantly abducted positioning ($p=0.076$). Partial spontaneous movement was observed in 4/5 biomolecule mice compared to 0/5 saline mice ($p=0.002$). Figure 2 displays representative 84-day video laryngoscopy images from the uninjured, saline, and biomolecule groups. Blinded videolaryngoscopy scoring across all animals is shown in Figure S2.

3.1.2 | Stimulated EMG

At 84 days, paired analysis of right (injured) versus left (uninjured) stimulated RLN EMG demonstrated no significant difference in AUC between sides in the biomolecule group ($p=0.177$), indicating restoration of symmetry between the injured and uninjured RLN. In contrast, the saline group showed a significant right-sided deficit at 84 days ($p=0.031$), consistent with persistent denervation of the injured RLN. No significant side difference was observed in the uninjured group ($p=0.325$). In the SN model, biomolecule and saline groups demonstrated significant asymmetry between injured and uninjured sides at 28 days ($p=0.001$ and $p=0.023$, respectively), which persisted at 84 days ($p=0.009$ and $p=0.038$), suggesting incomplete peripheral nerve recovery at these timepoints. Figure 3 shows (A) quantitative AUC differences between injured and uninjured RLNs at 84 days and (B) gross images of the regenerated RLN nerve band at harvest, which was present in all biomolecule-treated animals. Figure 4 shows stimulated SN EMG findings at 28 and 84 days, demonstrating persistent asymmetry between injured and uninjured sides across timepoints.

3.1.3 | Gait Analysis

All groups demonstrated a sharp decline in FCA ratio by day 3 following SN transection, confirming successful injury. Both the biomolecule and saline groups remained significantly below a ratio of 1.0 at all timepoints through 84 days, and significantly below the uninjured group throughout the observation period, indicating persistent gait asymmetry without meaningful recovery in either group. The biomolecule group showed a trend toward transient improvement on day 14 ($p=0.068$ vs. 1.0), representing the closest approach to symmetry observed across all timepoints, though this did not reach statistical significance. Figure 5 displays FCA ratio over time for the biomolecule, saline, and uninjured groups.

3.2 | Histopathological Analysis

3.2.1 | Immunohistochemistry

Figure 6 demonstrates immunofluorescent staining of the TA muscle at 84 days following RLN transection. Combined CHRNA1/SV2/NFL staining demonstrated qualitative differences in NMJ morphology and marker co-localization among groups. The biomolecule-treated group exhibited greater co-localization of presynaptic and postsynaptic markers compared to saline-treated animals, while normal controls demonstrated the most extensive staining pattern. These findings qualitatively suggest preservation of NMJ architecture following biomolecule treatment.

In the SN model, gross evaluation demonstrated improved continuity of the transected nerve in the biomolecule-treated group compared to saline. Nerve continuity was observed in 5/5 biomolecule-treated animals at 28 days and in 4/5 animals at 84 days. Corresponding NFL staining of gastrocnemius-associated nerve tissue qualitatively demonstrated reduced neuronal signal in saline and biomolecule-treated animals relative

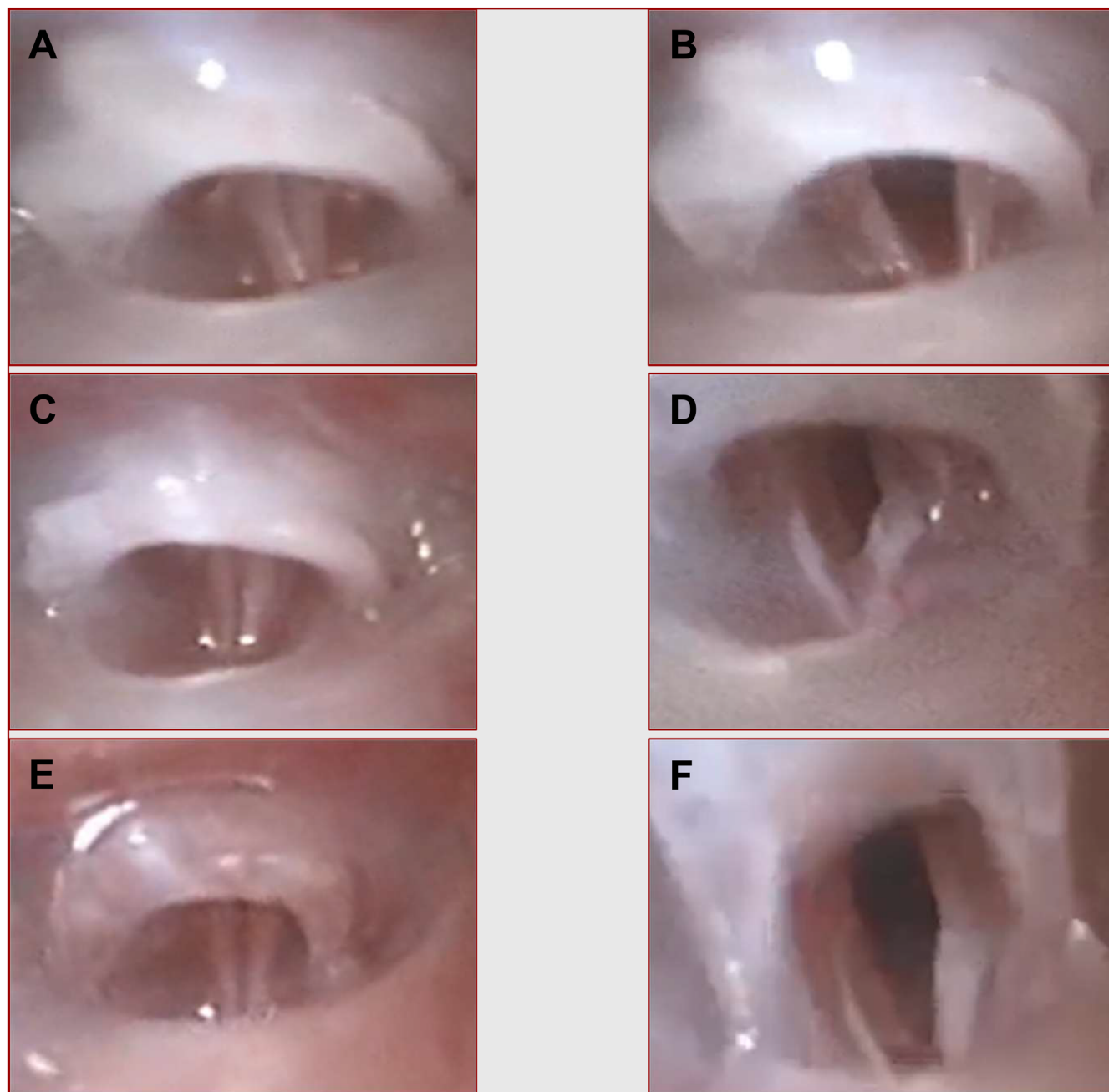


FIGURE 2 | Videolaryngoscopy assessment of bilateral vocal fold motion at 84 days following treatment. Images demonstrate adduction (left) and full abduction (right) of the vocal cords across groups: (A, B) uninjured; (C, D) saline-treated; and (E, F) biomolecule-treated. All treatment groups underwent right RLN transection and exhibited immediate postoperative immobility of the right VF. In uninjured animals, the right VF demonstrates smooth, symmetric abduction relative to the left, whereas saline-treated animals exhibit minimal right VF movement. Biomolecule-treated animals show partial restoration of right VF motion, with improved but incomplete symmetry compared to the contralateral side.

to uninjured, while the biomolecule group showed increased NFL-positive fibers compared to saline, suggesting partial axonal regeneration (Figure 7).

3.2.2 | Myofiber Diameter

At 84 days, the TA muscle myofiber diameter right-to-left ratio was significantly greater in the biomolecule group compared to the saline group ($p=0.015$), indicating muscle fiber symmetry with biomolecule treatment. The saline group demonstrated

a significantly reduced ratio compared to uninjured controls ($p=0.004$), consistent with atrophy of the injured right TA muscle. In contrast, the biomolecule group did not differ significantly from uninjured animals ($p=0.672$), suggesting restoration of myofiber integrity. Figure 8 and Table S2 display myofiber diameter ratios across groups.

In the SN model, neither 28- nor 84-day cohorts demonstrated significant differences in myofiber diameter symmetry between saline and biomolecule treatment groups ($p=0.522$, $p=0.508$). At 28 days, saline- ($0.839 \pm 0.077 \mu\text{m}$) and biomolecule-treated

A Right vs Left Stimulated EMG Symmetry at Time of Harvest Right RLN Transection (84 Days)

Colored dot = mean diff (Right – Left) $\pm$ 95% CI | Light dots = individual animals
Negative = Right (injured) < Left (uninjured) — injured side deficit

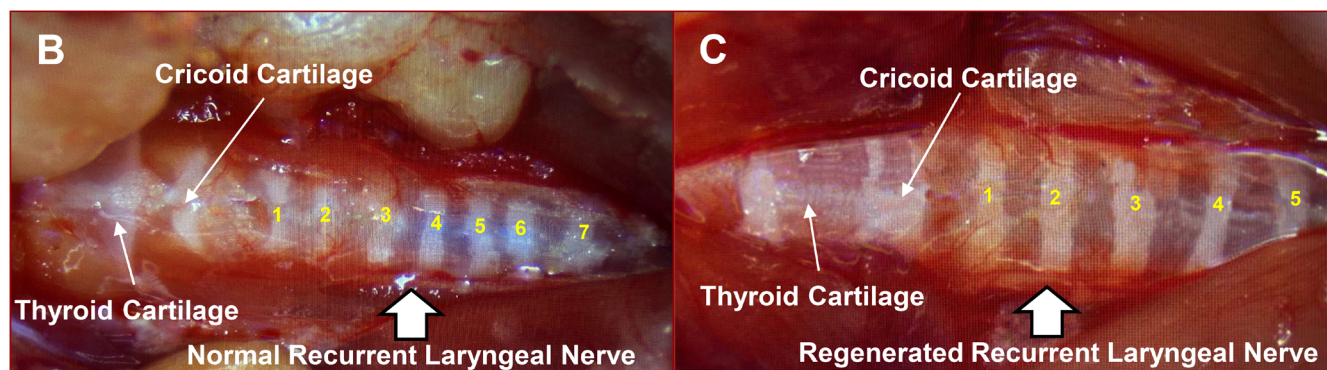
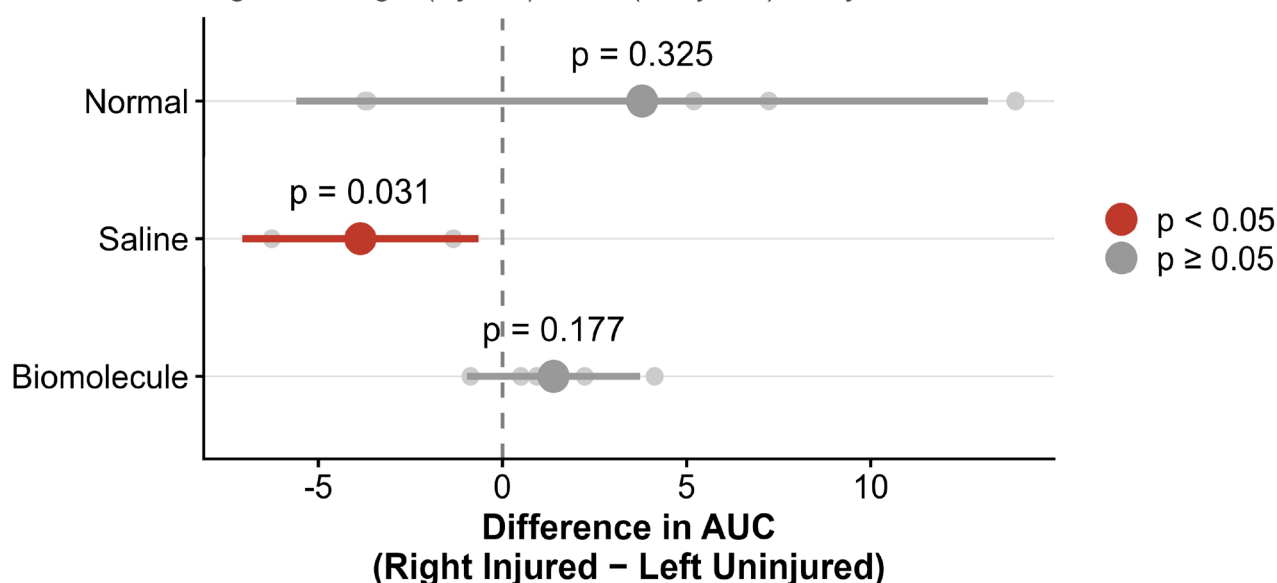


FIGURE 3 | RLN EMG symmetry and gross nerve appearance at 84 days following transection. (A) Comparison of right (injured) versus left (uninjured) stimulated RLN EMG activity at 84 days following transection, expressed as the difference in AUC between the right and left sides to assess bilateral symmetry. Small points represent individual animals, while large points indicate group means with lines denoting 95% confidence intervals. The dashed vertical line represents perfect symmetry between right and left EMG activity. Biomolecule-treated animals did not demonstrate significant asymmetry between the right and left sides, while saline exhibited persistent asymmetry, indicating restoration of bilateral symmetry with biomolecule treatment. (B) Representative gross image of the intact RLN in a uninjured mouse. (C) Representative gross image of the regenerated RLN 84 days following biomolecule treatment.

($0.844 \pm 0.090 \mu\text{m}$) animals demonstrated similar asymmetry compared to normal ($0.976 \pm 0.231 \mu\text{m}$). At 84 days, though not significant, saline-treated animals exhibited more asymmetry ($0.893 \pm 0.088 \mu\text{m}$) compared to the biomolecule ($0.938 \pm 0.049 \mu\text{m}$) and normal ($0.997 \pm 0.171 \mu\text{m}$) groups.

4 | Discussion

The primary aim of this study was to enhance TA adductor activity and muscle volume to improve glottic medialization in

unilateral VFP, rather than restore full motion. To this end, we evaluated the functional, histologic, and structural effects of Agrin/NGR1/ACh biomolecule treatment after nerve injury in both RLN and SN models. In the RLN model, biomolecule therapy supported functional reinnervation, NMJ preservation, and maintenance of neuromuscular structure through 84 days. In contrast, although early (28 day) structural and histologic signs of regeneration were observed after SN transection, changes did not translate into functional recovery, underscoring fundamental differences in requirements between the two systems [33–40].

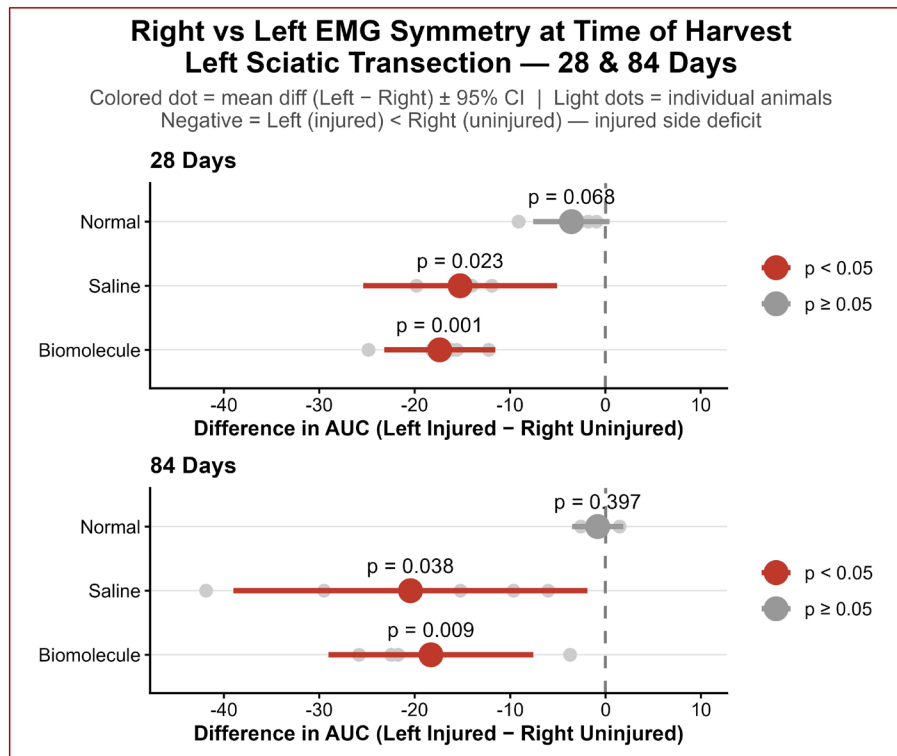


FIGURE 4 | SN EMG symmetry at 28 and 84 days following transection. Comparison of left (injured) versus right (uninjured) stimulated SN EMG activity, expressed as the difference in AUC between sides to assess bilateral symmetry. Small points represent individual animals, while large points indicate group means with lines denoting 95% confidence intervals. The dashed vertical line represents perfect symmetry between left and right EMG activity. Top panel: 28 days; bottom panel: 84 days. Persistent asymmetry was observed across saline and biomolecule-treated groups at both timepoints, **indicating no restoration of electrical symmetry** following SN injury.

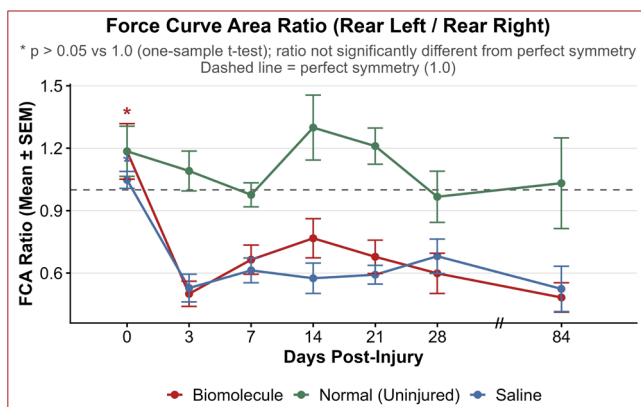


FIGURE 5 | Longitudinal functional recovery assessed by Bioseb kinetic weight-bearing (force curve area ratio, rear left/right). A ratio of 1.0 represents perfect symmetry between injured (left) and uninjured (right) hindlimbs, as indicated by the dashed horizontal line. Data are presented as group means with standard error of the mean across time points following injury, with an axis break between 28 and 84 days. Sample sizes reflect separate timepoint groups, with $n = 10$ through 28 days and $n = 5$ at all later timepoints. Asterisks indicate values not significantly different from 1.0 ($p > 0.05$, one-sample t -test). Biomolecule-treated animals demonstrated a nonsignificant trend toward improved symmetry at early timepoints but remained persistently asymmetric, similar to saline-treated animals, through 84 days. (At 14 days, Normal = top; Biomolecule = middle and Saline = bottom).

4.1 | RLN Model: Functional, Histologic, and Structural Recovery

Biomolecule treatment improved functional outcomes in the RLN model, demonstrated by restored VF adduction, enhanced response to RLN stimulation (Figure 2), and **improved EMG symmetry relative to saline controls** (Figure 3). Histologically, treated TA muscle showed greater co-localization of NF-L, SV2, and CHRNA1 (Figure 6), preserved RLN continuity, and reduced myofiber atrophy (Figure 8), consistent with maintenance or reformation of innervated NMJs and neuromuscular integrity. These effects likely reflect **complementary mechanisms**: biomolecule exposure can induce release of regenerative neurotrophic and angiogenic factors from treated muscle, while NRG1-ErbB signaling promotes Schwann cell repair phenotypes and axonal regeneration; Agrin-Lrp4-MuSK signaling stabilizes the postsynaptic apparatus, NRG1 enhances ACh receptor transcription, and ACh supports early synaptic maintenance during denervation. Together, these pathways create a microenvironment that stabilizes NMJs, supports targeted reinnervation, and improves RLN functional recovery [16–17, 20, 25–30].

The improved VF adduction in this model is clinically relevant, as restoration of adductor tone and glottic closure is a primary goal after unilateral RLN injury. The murine RLN, however, has a shorter regeneration distance and a simpler

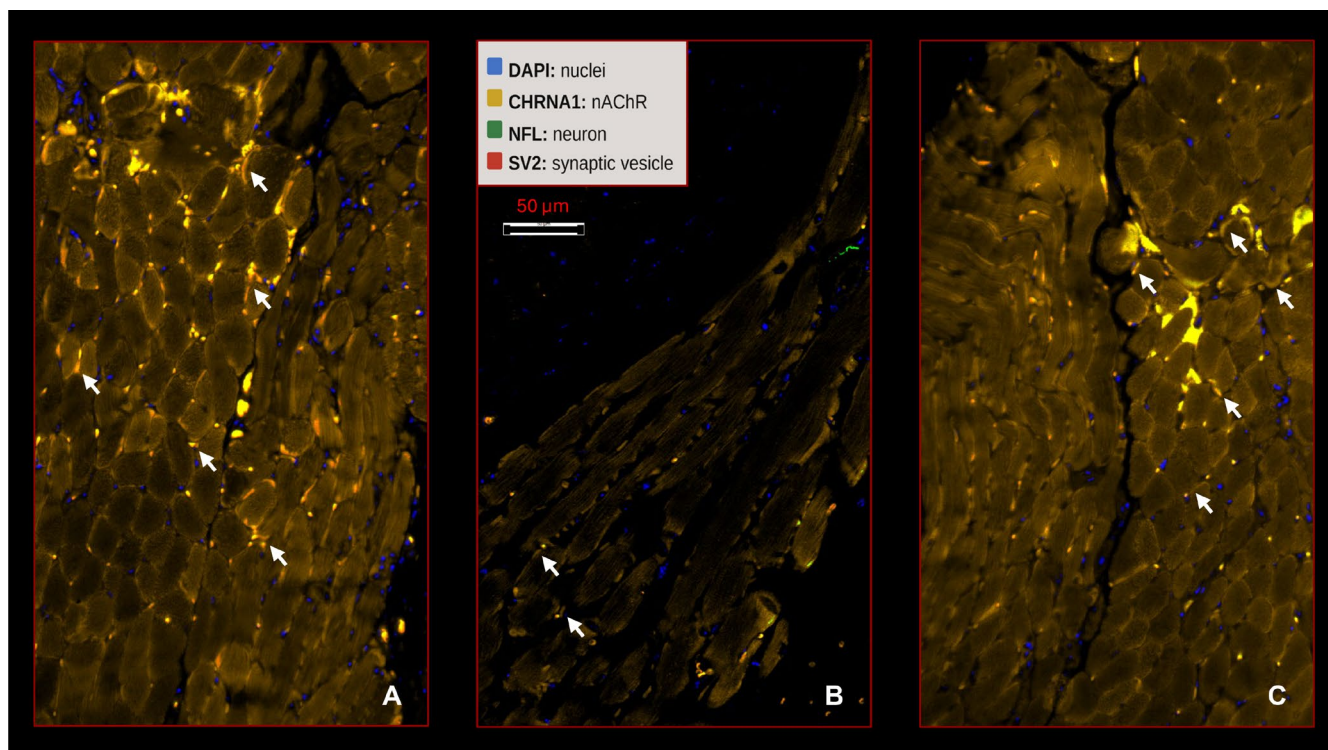


FIGURE 6 | Immunohistochemical assessment of neuromuscular junction morphology following RLN injury. Representative merged immunofluorescence images of the right (injured) TA muscle at 84 days following RLN transection from (A) uninjured mice, (B) saline-treated mice, and (C) biomolecule-treated mice. Sections were stained for CHRNA1 (orange), neurofilament light chain (NFL; green), synaptic vesicle protein 2 (SV2; red), and DAPI (blue). Arrows indicate regions of CHRNA1 hyperintensity adjacent to NFL- and SV2-positive structures, consistent with innervated neuromuscular junctions. Compared with saline-treated animals, biomolecule-treated animals demonstrated qualitatively greater co-localization of neuromuscular junction markers.

intrinsic laryngeal motor pool than the human larynx, features that likely facilitate recovery in this model [42, 43]. In humans, even partial axonal injury commonly produces synkinetic or dyskinetic reinnervation of adductor and abductor muscles, disrupting coordinated phonation and respiration [7, 11]. Synkinesis was not directly quantified in the present study, so the extent to which localized NMJ stabilization mitigates aberrant motor reinnervation cannot be inferred from these data. Future work in larger-animal models and human studies will be necessary to determine how such biologic strategies can optimize functional outcomes.

4.2 | SN Model: Lack of Functional Recovery Despite Partial Histologic and Structural Reinnervation

In the SN model, both biomolecule-treated and saline-treated animals exhibited persistent functional deficits, with no improvement in EMG symmetry (Figure 4) or gait parameters (Figure 5). Nonetheless, biomolecule-treated nerves demonstrated early structural indicators of regeneration, including gross continuity, increased NFL-positive fibers, and partial preservation of myofiber architecture (Figure 7).

This dissociation between structure and function reflects the markedly different regenerative demands of the two models. With the RLN being a motor nerve, the primary recovery

challenge is misdirected motor reinnervation leading to inappropriate central input (abduction rather than adduction). By contrast, the SN is a large-diameter mixed sensory-motor nerve innervating multiple muscles; sensory and motor fibers from different fascicles attempt to regenerate simultaneously, and trophic factor and Schwann cell support alone cannot resolve this complex intraneural anatomy, for which precise fascicular alignment would be required. Because of these anatomic obstacles, to our knowledge no single injectable has consistently restored early (28-day) nerve continuity in an unrepaired SN injury model. In this context, Agrin/NGR1/ACh likely promotes early events such as Bands of Büngner formation and Schwann cell repair phenotypes, and the observation that nerve ends reconnected anatomically and contained viable NFL-positive axons is notable, suggesting that this biomolecular strategy can drive early structural regeneration even in a highly unfavorable mechanical and anatomic environment. The reduced nerve continuity at 84 days indicates that early bridging tissue failed to mature into stable functional connections, and future models will therefore need to integrate structural support for the nerve and limb together with repeat dosing.

4.3 | Limitations

This study is limited by using a small animal model and modest sample size. Electrophysiologic assessment in the RLN

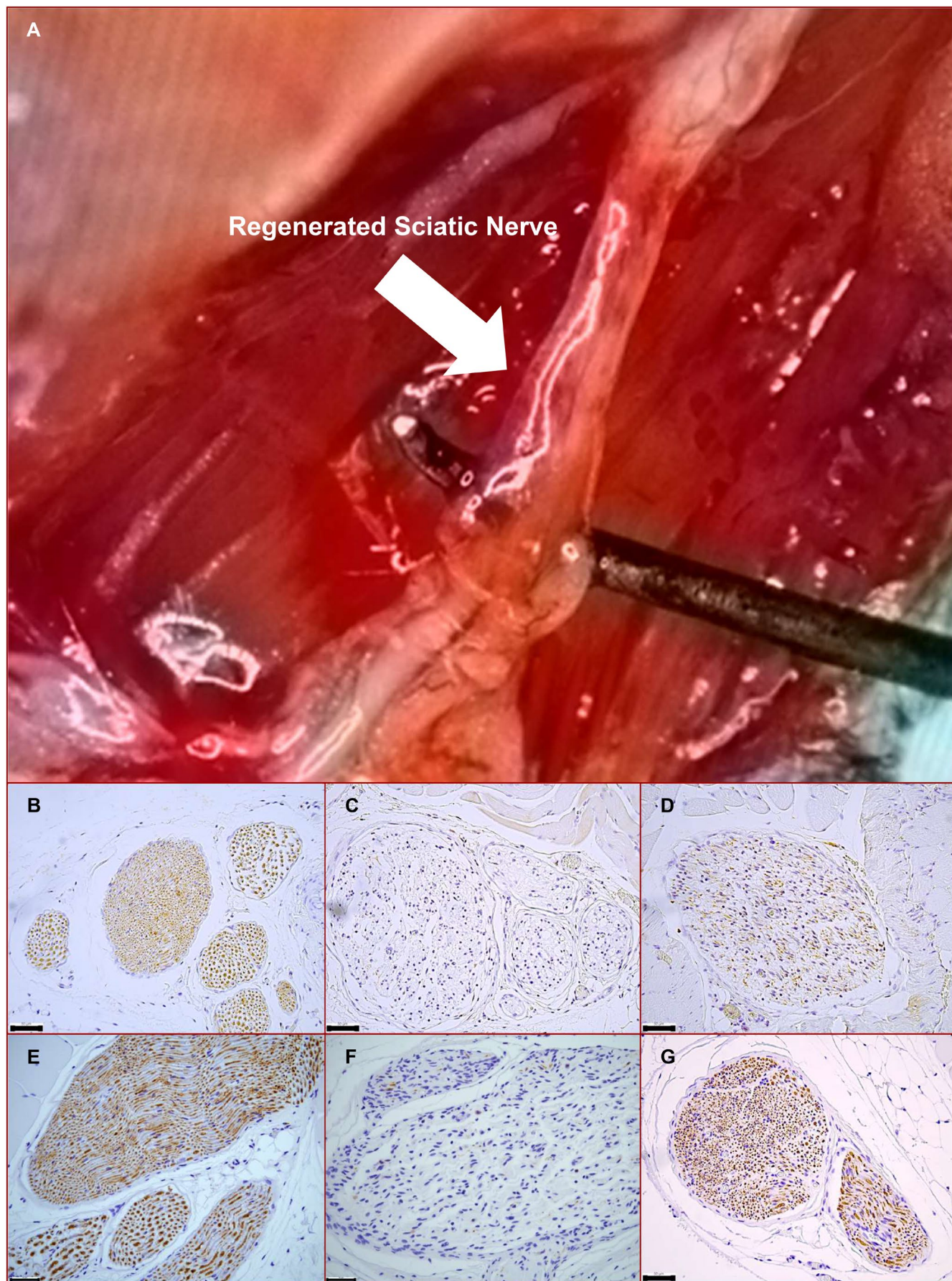


FIGURE 7 | Legend on next page.

FIGURE 7 | SN regeneration and neuronal staining following treatment. (A) Representative gross image of regenerated SN tissue in the biomolecule-treated group at harvest, demonstrating a continuous nerve band at the site of prior transection. (B–D) NFL staining of nerve sections within harvested gastrocnemius muscle sections at 28 days taken on the Leica DM2500 microscope from (B) uninjured mice, (C) saline-treated mice, and (D) biomolecule-treated mice. (E–G) NFL staining of corresponding nerve tissue sections at 84 days from (E) uninjured mice, (F) saline-treated mice, and (G) biomolecule-treated mice. Immunopositivity (brown) indicates the presence of neuronal structures. Biomolecule-treated samples demonstrated greater NFL positive staining than saline but less than uninjured at 28 days, with increased staining at 84 days. Saline-treated animals showed minimal to no staining at either timepoint. These findings are consistent with enhanced axonal regeneration in biomolecule-treated animals.

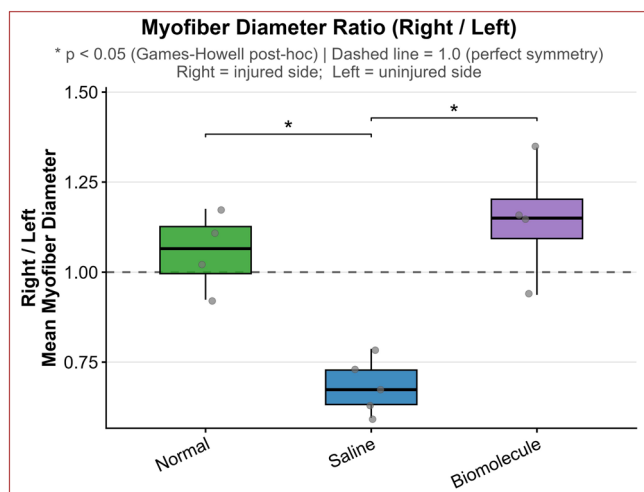


FIGURE 8 | Myofiber diameter ratio of the TA muscle at 84 days following RLN transection. Myofiber diameter is expressed as the ratio of right (injured) to left (uninjured) muscle, with the dashed line indicating perfect symmetry (1.0). Light points represent individual animals, while the central horizontal line indicates the group mean with 95% confidence intervals. Asterisks denote statistically significant differences between groups ($p < 0.05$, Games–Howell post hoc). Biomolecule-treated animals demonstrated significantly greater myofiber diameter ratios compared to saline and were not significantly different from uninjured controls, indicating preservation of myofiber symmetry.

model was limited to the TA muscle. Attempts to obtain posterior cricoarytenoid (PCA) EMG recordings produced mucosal edema/airway compromise so synkinetic reinnervation was not assessed. Future studies in larger-animal models will incorporate PCA EMG and VF motion analysis to characterize synkinesis and laryngeal motion recovery. The treatment protocol was also limited to a single administration at the time of injury, and future studies are needed to evaluate the potential benefits of repeated dosing. NMJs and NFL were not quantitatively assessed, which is planned for future studies. Finally, further evaluation in larger animal models and human studies will be necessary to determine the clinical applicability of this treatment.

5 | Conclusions

Biomolecule therapy consisting of Agrin, NRG1, and Ach demonstrates strong potential as an injectable treatment for RLN injury, promoting functional reinnervation, NMJ preservation, and maintenance of muscle integrity through 84 days.

Although early structural regeneration was observed in the SN model, these findings did not translate into functional recovery, likely reflecting the greater demands of long-distance axonal regeneration compared with the more localized, target-specific reinnervation of the RLN. Future work will incorporate sustained-release delivery, repeat dosing, and structural support/less severe injury paradigms to better assess outcomes in peripheral nerve injury models.

Acknowledgments

We would like to thank the IU School of Medicine Histology Core and IU Laboratory Animal Resource Center that housed our C57/BL6 mice utilized in the study. The research was funded by IUSM Department of Otolaryngology—HNS.

Funding

The research was funded by the IUSM Department of Otolaryngology—HNS, 5R01DC019632-05.

Ethics Statement

This study was performed in accordance with the PHS Policy on Humane Care and Use of Laboratory Animals, the NIH Guide for the Care and Use of Laboratory Animals, and the Animal Welfare Act (7 U.S.C. et seq.); the Indiana University Institutional Animal Care and Use Committee (IACUC) approved the animal use protocol (23150; approved February 9, 2024).

Conflicts of Interest

Dr. Stacey Halum is the sole owner and Chief Executive Officer of Vibrant Therapeutics Inc., a startup company developing and commercializing neuromuscular therapies. Patrick Finnegan is an employee of the Indiana University School of Medicine (IUSM) and has no ownership, equity, executive role, or financial interest in Vibrant Therapeutics Inc. Dr. Halum and Patrick R. Finnegan are coinventors on an IUSM-held patent relevant to this work; all patent rights and any associated royalties are assigned to and administered by IUSM under standard institutional policy. To mitigate potential bias, the following oversight and blinding mechanisms were implemented: (1) all surgical procedures, in vivo functional assessments (transoral laryngoscopy, EMG, gait analysis), histologic processing, imaging, and quantitative analyses were performed by researchers with no affiliation to Vibrant Therapeutics Inc.; (2) all subjective and semi-quantitative outcome assessments (vocal fold motion scoring, NMJ co-localization, gross nerve evaluation) were conducted by blinded reviewers; and (3) treatment group codes were not unblinded until after all outcome data were finalized. The other authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. D. O. Francis, A. E. Sherman, K. L. Hovis, et al., "Life Experience of Patients With Unilateral Vocal Fold Paralysis," *JAMA Otolaryngology—Head & Neck Surgery* 144, no. 5 (2018): 433–439, <https://doi.org/10.1001/jamaoto.2018.0067>.
2. J. P. Sand, A. M. Park, N. Bhatt, et al., "Comparison of Conventional, Revascularized, and Bioengineered Methods of Recurrent Laryngeal Nerve Reconstruction," *JAMA Otolaryngology—Head & Neck Surgery* 142, no. 6 (2016): 526–532.
3. A. Mok, J. Allen, M. M. Haney, et al., "A Surgical Mouse Model for Advancing Laryngeal Nerve Regeneration Strategies," *Dysphagia* 35, no. 3 (2020): 419–437, <https://doi.org/10.1007/s00455-019-10045-6>.
4. E. O. Shay, R. A. Morrison, L. Zhang, et al., "Scaffold-Forming Collagen and Motor-Endplate Expressing Muscle Cells for Porcine Laryngoplasty," *Laryngoscope* 134, no. 12 (2024): 4988–4997, <https://doi.org/10.1002/lary.31642>.
5. R. R. Patel, S. N. Awan, J. Barkmeier-Kraemer, et al., "Recommended Protocols for Instrumental Assessment of Voice: American Speech-Language-Hearing Association Expert Panel to Develop a Protocol for Instrumental Assessment of Vocal Function," *American Journal of Speech-Language Pathology* 27, no. 3 (2018): 887–905, https://doi.org/10.1044/2018_AJSLP-17-0009.
6. D. H. Lee, S. Y. Lee, M. Lee, et al., "Natural Course of Unilateral Vocal Fold Paralysis and Optimal Timing of Permanent Treatment," *JAMA Otolaryngology—Head & Neck Surgery* 146, no. 1 (2020): 30–35, <https://doi.org/10.1001/jamaoto.2019.3072>.
7. G. Foerster and A. H. Mueller, "PCA Atrophy and Synkinesis as the Main Factors for Persistent Vocal Fold Immobility in RLN Paralysis," *Laryngoscope* 131, no. 4 (2021): E1244–E1248, <https://doi.org/10.1002/lary.29195>.
8. B. Tessema, R. M. Roark, M. J. Pitman, P. Weissbrod, S. Sharma, and S. D. Schaefer, "Observations of Recurrent Laryngeal Nerve Injury and Recovery Using a Rat Model," *Laryngoscope* 119, no. 8 (2009): 1644–1651, <https://doi.org/10.1002/lary.20293>.
9. C. B. Simpson, E. J. Cheung, and C. J. Jackson, "Vocal Fold Paresis: Clinical and Electrophysiologic Features in a Tertiary Laryngology Practice," *Journal of Voice* 23, no. 3 (2009): 396–398, <https://doi.org/10.1016/j.jvoice.2007.10.011>.
10. R. Philips, K. Chorath, B. DeSilva, L. A. Forrest, C. B. Simpson, and L. Matrk, "Multi-Institutional Evaluation of Medialization Laryngoplasty in the Elderly," *Otolaryngology and Head and Neck Surgery* 160, no. 5 (2019): 876–884, <https://doi.org/10.1177/0194599818817762>.
11. Y. C. Pei, W. H. Chang, H. F. Chuang, C. F. Chang, and T. J. Fang, "Implications of Synkinesis in Unilateral Vocal Fold Paralysis," *Otolaryngology and Head and Neck Surgery* 157, no. 6 (2017): 1017–1024, <https://doi.org/10.1177/0194599817721688>.
12. O. Guntinas-Lichius, A. Irintchev, M. Streppel, et al., "Factors Limiting Motor Recovery After Facial Nerve Transection in the Rat: Combined Structural and Functional Analyses," *European Journal of Neuroscience* 21, no. 2 (2005): 391–402, <https://doi.org/10.1111/j.1460-9568.2005.03877.x>.
13. S. L. Halum, K. Bijangi-Vishehsaraei, M. R. Saadatzaadeh, and B. R. McRae, "Differences in Laryngeal Neurotrophic Factor Gene Expression After Recurrent Laryngeal Nerve and Vagus Nerve Injuries," *Annals of Otolaryngology, Rhinology, and Laryngology* 122, no. 10 (2013): 653–663.
14. S. L. Halum, K. Bijangi-Vishehsaraei, H. Zhang, J. Sowinski, and M. C. Bottino, "Stem Cell-Derived Tissue-Engineered Constructs for Hemilaryngeal Reconstruction," *Annals of Otolaryngology, Rhinology, and Laryngology* 123, no. 2 (2014): 124–134, <https://doi.org/10.1177/0003489414523709>.
15. R. C. Paniello, S. Brookes, N. K. Bhatt, K. Bijangi-Vishehsaraei, H. Zhang, and S. Halum, "Improved Adductor Function After Canine Recurrent Laryngeal Nerve Injury and Repair Using Muscle Progenitor Cells," *Laryngoscope* 128, no. 7 (2018): E241–E246, <https://doi.org/10.1002/lary.26992>.
16. S. L. Kaefer, L. Zhang, R. A. Morrison, et al., "Early Changes in Porcine Larynges Following Injection of Motor-Endplate Expressing Muscle Cells for the Treatment of Unilateral Vocal Fold Paralysis," *Laryngoscope* 134, no. 1 (2024): 272–282, <https://doi.org/10.1002/lary.30868>.
17. S. L. Kaefer, L. Zhang, S. Brookes, R. A. Morrison, S. Voytik-Harbin, and S. Halum, "Optimizing Transport Methods to Preserve Function of Self-Innervating Muscle Cells for Laryngeal Injection," *Laryngoscope Investigative Otolaryngology* 9, no. 6 (2024): e1259.
18. X. Vega-Cordova, N. M. Cosenza, R. H. Helfert, and G. E. Woodson, "Neurotrophin Expression of Laryngeal Muscles in Response to Recurrent Laryngeal Nerve Transection," *Laryngoscope* 120, no. 8 (2010): 1591–1596.
19. T. Gordon, "Peripheral Nerve Regeneration and Muscle Reinnervation," *International Journal of Molecular Sciences* 21, no. 22 (2020): 8652, <https://doi.org/10.3390/ijms21228652>.
20. B. A. Cisterna, A. A. Vargas, C. Puebla, et al., "Active Acetylcholine Receptors Prevent the Atrophy of Skeletal Muscles and Favor Reinnervation," *Nature Communications* 11, no. 1 (2020): 1073, <https://doi.org/10.1038/s41467-019-14063-8>.
21. S. Rink-Notzon, J. Reuscher, L. Wollny, et al., "Appropriate Dosage, Timing, and Site of Intramuscular Injections of Brain-Derived Neurotrophic Factor (BDNF) Promote Motor Recovery After Facial Nerve Injury in Rats," *Muscle & Nerve* 69, no. 4 (2024): 490–497, <https://doi.org/10.1002/mus.28051>.
22. S. T. Ngo, R. N. Cole, N. Sunn, W. D. Phillips, and P. G. Noakes, "Neuregulin-1 Potentiates Agrin-Induced Acetylcholine Receptor Clustering Through Muscle-Specific Kinase Phosphorylation," *Journal of Cell Science* 125, no. Pt 6 (2012): 1531–1543.
23. B. G. X. Zhang, A. F. Quigley, J. L. Bourke, et al., "Combination of Agrin and Laminin Increase Acetylcholine Receptor Clustering and Enhance Functional Neuromuscular Junction Formation In Vitro," *Developmental Neurobiology* 76, no. 5 (2016): 551–565, <https://doi.org/10.1002/dneu.22331>.
24. A. J. Rosko, R. A. Kupfer, S. S. Oh, C. T. Haring, E. L. Feldman, and N. D. Hogikyan, "Immunohistologic Analysis of Spontaneous Recurrent Laryngeal Nerve Reinnervation in a Rat Model," *Laryngoscope* 128, no. 3 (2018): E117–E122, <https://doi.org/10.1002/lary.27004>.
25. K. S. Grossmann, H. Wende, F. E. Paul, et al., "The Tyrosine Phosphatase Shp2 (PTPN11) Directs Neuregulin-1/ErbB Signaling Throughout Schwann Cell Development," *Proceedings of the National Academy of Sciences of the United States of America* 106, no. 39 (2009): 16704–16709, <https://doi.org/10.1073/pnas.0904336106>.
26. S. Hettwer, S. Lin, S. Kucsera, et al., "Injection of a Soluble Fragment of Neural Agrin (NT-1654) Considerably Improves the Muscle Pathology Caused by the Disassembly of the Neuromuscular Junction," *PLoS One* 9, no. 2 (2014): e88739, <https://doi.org/10.1371/journal.pone.0088739>.
27. Q. Li and J. A. Loeb, "Neuregulin-Heparan-Sulfate Proteoglycan Interactions Produce Sustained erbB Receptor Activation Required for the Induction of Acetylcholine Receptors in Muscle," *Journal of Biological Chemistry* 276, no. 41 (2001): 38068–38075.
28. J. F. Yang, G. Cao, S. Koirala, L. V. Reddy, and C. P. Ko, "Schwann Cells Express Active Agrin and Enhance Aggregation of Acetylcholine Receptors on Muscle Fibers," *Journal of Neuroscience* 21, no. 24 (2001): 9572–9584, <https://doi.org/10.1523/JNEUROSCI.21-24-09572.2001>.
29. G. V. Michailov, M. W. Sereda, B. G. Brinkmann, et al., "Axonal Neuregulin-1 Regulates Myelin Sheath Thickness," *Science* 304, no. 5671 (2004): 700–703, <https://doi.org/10.1126/science.1095862>.

30. Q. Min, D. B. Parkinson, and X. P. Dun, "Migrating Schwann Cells Direct Axon Regeneration Within the Peripheral Nerve Bridge," *Glia* 69, no. 2 (2021): 235–254, <https://doi.org/10.1002/glia.23892>.
31. Y. Ma, L. Dong, D. Zhou, et al., "Extracellular Vesicles From Human Umbilical Cord Mesenchymal Stem Cells Improve Nerve Regeneration After Sciatic Nerve Transection in Rats," *Journal of Cellular and Molecular Medicine* 23, no. 4 (2019): 2822–2835, <https://doi.org/10.1111/jcmm.14190>.
32. B. B. Aynehchi, E. D. McCoul, and K. Sundaram, "Systematic Review of Laryngeal Reinnervation Techniques," *Otolaryngology—Head and Neck Surgery* 143, no. 6 (2010): 749–759, <https://doi.org/10.1016/j.otohns.2010.09.031>.
33. M. B. Montalbano, I. Hernández-Morato, L. Tian, et al., "Recurrent Laryngeal Nerve Reinnervation in Rats Posttransection: Neurotrophic Factor Expression Over Time," *Otolaryngology and Head and Neck Surgery* 161, no. 1 (2019): 111–117, <https://doi.org/10.1177/0194599819831289>.
34. A. Tadikonda, S. Anekal, L. W. Chen, et al., "Novel Biomolecule-Infused Gelatin Injectable for Treatment of Recurrent Laryngeal Nerve Injury," *Laryngoscope* 135, no. 12 (2025): 4781–4792, <https://doi.org/10.1002/lary.32459>.
35. A. Švejdová, J. Dršata, J. Mejzlík, et al., "Injection Laryngoplasty With Hyaluronic Acid for Glottal Insufficiency in Unilateral Vocal Fold Paralysis: A Systematic Review of the Literature," *European Archives of Oto-Rhino-Laryngology* 279, no. 11 (2022): 5071–5079, <https://doi.org/10.1007/s00405-022-07437-0>.
36. H. Takeya, S. Itai, H. Kimura, et al., "Schwann Cell-Encapsulated Chitosan-Collagen Hydrogel Nerve Conduit Promotes Peripheral Nerve Regeneration in Rodent Sciatic Nerve Defect Models," *Scientific Reports* 13, no. 1 (2023): 11932, <https://doi.org/10.1038/s41598-023-39141-2>.
37. T. C. Olsen, J. S. LaGuardia, D. R. Chen, et al., "Influencing Factors and Repair Advancements in Rodent Models of Peripheral Nerve Regeneration," *Regenerative Medicine* 19, no. 11 (2024): 561–577, <https://doi.org/10.1080/17460751.2024.2405318>.
38. F. Sousa, M. Santos, S. Azevedo, et al., "Unilateral Vocal Fold Paralysis and Voice Therapy: Predictors of Long-Term Quality of Life," *Cureus* 15, no. 1 (2023): e34078, <https://doi.org/10.7759/cureus.34078>.
39. Y. Kumai, T. Ito, A. Matsukawa, and E. Yumoto, "Effects of Denervation on Neuromuscular Junctions in the Thyroarytenoid Muscle," *Laryngoscope* 115, no. 10 (2005): 1869–1872, <https://doi.org/10.1097/01.mlg.0000177076.33294.89>.
40. S. H. Ko and L. Chen, "Mouse Sciatic Nerve Transection/Resuture Procedure for Studying Nerve Repair and Functional Recovery," *Journal of Visualized Experiments* 227 (2026): 68724, <https://doi.org/10.3791/68724>.
41. S. E. Mackinnon, A. R. Hudson, and D. A. Hunter, "Histologic Assessment of Nerve Regeneration in the Rat," *Plastic and Reconstructive Surgery* 75, no. 3 (1985): 384–388, <https://doi.org/10.1097/00006534-198503000-00014>.
42. P. Weissbrod, M. J. Pitman, S. Sharma, A. Bender, and S. D. Schaefer, "Quantity and Three-Dimensional Position of the Recurrent and Superior Laryngeal Nerve Lower Motor Neurons in a Rat Model," *Annals of Otology, Rhinology, and Laryngology* 120, no. 11 (2011): 761–768, <https://doi.org/10.1177/000348941112001111>.
43. R. Tiago, P. Pontes, and O. C. do Brasil, "Age-Related Changes in Human Laryngeal Nerves," *Otolaryngology—Head and Neck Surgery* 136, no. 5 (2007): 747–751, <https://doi.org/10.1016/j.otohns.2006.11.054>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Representative sectioning and staining of mouse larynx and gastrocnemius muscle. (A) Coronal

section of mouse larynx stained with hematoxylin and eosin-Y for myofiber diameter quantitation. (B) Coronal section of mouse gastrocnemius stained with hematoxylin and eosin-Y for myofiber diameter quantitation. (C, D) Representative image of minimum Feret's myofiber diameter measurement in mouse TA and gastrocnemius muscles, respectively. **Figure S2:** Blinded videolaryngoscopy scoring at 84 days following RLN transection. Dot plots display individual animal scores with group means (black bars) for spontaneous VF movement (left), vocal cord position (middle), and right (injured) RLN stimulation-evoked responses (right) across uninjured, saline, and biomolecule-treated groups. All assessments were performed in a blinded fashion. Biomolecule-treated animals demonstrated significantly improved right RLN stimulation responses compared to saline, with trends toward improvement in spontaneous movement and VF position. **Data S1:** Supporting Information: Expanded Methods, Results, and Discussion. **Table S1:** Distribution of animals across experimental groups and harvest time points. Separate cohorts of animals were used for RLN and SN experiments. Independent SN cohorts were assigned to 28-day and 84-day harvest time points. RLN cohort: $n = 15$ at 84 days; SN cohorts: $n = 15$ at 28 days and $n = 15$ at 84 days; total $n = 45$. **Table S2:** Myofiber minimum Feret diameter by group and injury model. Values are mean $\pm$ SD of minimum Feret (minFeret) diameter (μm) measured from myofiber sections in C57BL/6 mice. Injured/uninjured (I/U) ratio is computed within each animal as the injured-limb mean divided by the contralateral uninjured-limb mean; a ratio of 1.0 indicates no atrophy. For the RLN model, injured refers to the right RLN while uninjured refers to the left; for the SN model, the left leg is injured and the right is uninjured. Within each model and timepoint, group ratios were compared by Welch ANOVA with Games-Howell post hoc; tabulated p values are Games-Howell comparisons relative to the Normal group. The RLN ANOVA was significant ($W = 22.32$, $df = 2$, 5.49 ; $p = 0.002$). The SN ANOVA was nonsignificant at both timepoints (28-day: $W = 0.745$, $p = 0.508$; 84-day: $W = 0.730$, $p = 0.522$), so no SN pairwise comparison reached significance. $n = 4$ –5 animals per group. * $p < 0.05$, ** $p < 0.01$, ns = not significant. SD, standard deviation; μm , micrometers.